2018

I)

Q a) Show that $\int_0^\infty \frac{\sin at}{t}\, dt = \frac{\pi}{2}$

→ Let $f(t) = \sin at$

$L\{f(t)\} = L\{\sin at\}$

$f(s) = \frac{a}{s^2+a^2}$

$f(u) = \frac{a}{u^2+a^2}$

we know that

$$\int_0^\infty \frac{F(t)}{t}\, dt = \int_0^\infty f(u)\, du$$

or $$\int_0^\infty \frac{\sin at}{t}\, dt = \int_0^\infty \frac{a}{u^2+a^2}\, du$$

$$= a\int_0^\infty \frac{1}{u^2+a^2}\, du$$

$$= \cancel{a}\cdot\frac{1}{\cancel{a}}\left[\tan^{-1}\frac{u}{a}\right]_0^\infty$$

$$= \tan^{-1}\infty - \tan^{-1}0$$

$$= \frac{\pi}{2}$$

transform of $\cot^{-1}\left(\frac{s+3}{2}\right)$

Let

End Semester Examination winter 2018

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LS

Q 1

c) Find the Laplace transform of the function

$$f(t) = \begin{cases} 2 & 0<t<\pi \\ 0 & \pi<t<2\pi \\ \sin t & t>2\pi \end{cases}$$

$\int_0^{\pi}$ $\int_{\pi}^{2\pi}$ $\int_{\pi}^{2\pi}$

i) The function F(t) can be expressed as under.

$$F(t) = 2\{u(t)-u(t-\pi)\} + (0)\{u(t-\pi)-u(t-2\pi)\} + \sin t\{u(t-\pi)\}$$

$$= 2u(t) - 2u(t-\pi) + \sin t\, u(t-2\pi)$$

$$\therefore L\{f(t)\} = L\{2u(t)\} - 2u(t-\pi) + \sin t\, u(t-2\pi)\}$$

$$= 2L\{u(t)\} - 2L\{u(t-\pi)\} + L\{\sin(t-2\pi)\, u(t-2\pi)\}$$

$$= 2\left\{\frac{1}{s}\right\} - 2\left\{\frac{e^{-\pi s}}{s}\right\} + e^{-2\pi s} L\{\sin t\}$$

$$= \frac{2}{s} - \frac{2e^{-\pi s}}{s} + \frac{e^{-2\pi s}}{s^2+1}$$

$$\left[\therefore L\{u(t-a)\} = \frac{e^{-as}}{s} \quad \text{and} \quad L\{f(t-a)\,u(t-a)\} = e^{-as} L\{f(t)\}\right]$$

$$-\frac{}{2}.$$

Q2. Find the inverse Laplace transform of $\cot^{-1}\left(\frac{s+3}{2}\right)$

1) $\Rightarrow$ Let $f(s) = \cot^{-1}\left(\frac{s+3}{2}\right)$

$$\frac{df(s)}{ds} = \frac{d}{ds}\left[\cot^{-1}\left(\frac{s+3}{2}\right)\right] = \frac{-1}{1+\left(\frac{s+3}{2}\right)^2}\left(\frac{1}{2}\right) = \frac{-1}{1+\left(\frac{(s+3)^2}{4}\right)}\cdot\frac{1}{2}$$

$$\frac{df(s)}{ds} = \frac{-1}{\frac{4+(s+3)^2}{\cancel{4}\,2}}\cdot\frac{1}{\cancel{2}} = \frac{-2}{(s+3)^2+2^2}$$

$$L^{-1}\left\{\frac{d}{ds}f(s)\right\} = -L^{-1}\left\{\frac{2}{(s+3)^2+2^2}\right\}$$

$$-tf(t) = e^{-3t}\sin 2t$$

$$\boxed{f(t) = \frac{e^{-3t}\sin 2t}{t}}$$

Q2. b) By convolution theorem find inverse Laplace transform of $\frac{s}{(s^2+1)(s^2+4)}$

$\rightarrow \bar{f}_1(s) = \frac{1}{s^2+1} \quad L^{-1}\{\bar{f}_1(s)\} = f_1(t) = L^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t$

$\bar{F}_2(s) = \frac{s}{s^2+4} \quad L^{-1}\{\bar{f}_2(s)\} = f_2(t) = L^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos 2t$

Now $f_1(t) = \sin t \longrightarrow f_1(u) = \sin u$

$f_2(t) = \cos 2t \longrightarrow f_2(t-u) = \cos 2(t-u) = \cos(2t-2u)$

By convolution theorem we have.

$L^{-1}\{\bar{f}(s)\} = L^{-1}\{\bar{f}_1(s)\bar{f}_2(s)\} = \int_0^t f_1(u) f_2(t-u)\,du = \int_0^t \sin u \cos(2t-2u)\,du$

$= \frac{1}{2}\int_0^t 2\sin u \cos(2t-2u)\,du = \frac{1}{2}\int_0^t [\sin(u+2t-2u) + \sin(u-2t+2u)]\,du$

$= \frac{1}{2}\int [\sin(2t-u) + \sin(3u-2t)\,du = \frac{1}{2}\int_0^t [\sin(u+2t-2u) + \sin(u-2t+2u)\,du$

$= \frac{1}{2}\left\{\left[\frac{-\cos(2t-u)}{-1}\right]_0^t + \left[\frac{-\cos(3u-2t)}{3}\right]_0^t\right\}$

$= \frac{1}{2}\left[\frac{\cos(2t-t)}{1} - \cos(2t-0)\right] + \left[\frac{-\cos(3t-2t)}{3} + \frac{\cos(0-2t)}{3}\right]$

$= \frac{1}{2}\left\{[\cos t - \cos 2t] + \left[-\frac{\cos t}{3} + \frac{\cos 2t}{3}\right]\right\}$

$= \cos t\left(\frac{1}{2}-\frac{1}{3}\right) - \cos 2t\left(\frac{1}{2}-\frac{1}{3}\right) = \cos t\left(\frac{6-2}{12}\right) - \cos 2t\left(\frac{6-2}{12}\right)$

$= \cos t\left(\frac{4}{12}\right) - \cos 2t\left(\frac{4}{12}\right) = \cos t\left(\frac{1}{3}\right) - \cos 2t\left(\frac{1}{3}\right) = \frac{1}{3}[\cos t - \cos 2t]$

Q2.

c) By Laplace tranform method solve the following simultaneous equations.

$$\frac{dx}{dt} - y = e^t \ : \ \frac{dy}{dt} + x = \sin t \ : \quad \text{given that } x(0) = 1, \ y(0) = 0.$$

→ The given simultaneous differential equation are

$$x'(t) - y(t) = e^t \quad \text{——} (1)$$

$$\text{where} \quad x(0) = 1 \quad y(0) = 0 \text{——} (3)$$

$$x(t) - y'(t) = \sin t \quad \text{——} (2)$$

Taking the Laplace transform of eqn (1) & (2) we have

$$s\bar{x}(s) - x(0) - \bar{y}(s) = \frac{1}{s-1}$$

$$\bar{x}(s) + s\bar{y}(s) - y(0) = \frac{1}{s^2+1}$$

where $x(0) = 1$, $y(0) = 0$

$$s\bar{x}(s) - \bar{y}(s) = \frac{1}{s-1} + \frac{1}{1} = \frac{1+s-1}{s-1} = \frac{s}{s-1} \quad \text{——} (4)$$

$$\bar{x}(s) + s\bar{y}(s) = \frac{1}{s^2+1} \quad \text{——} (5)$$

By eliminate $\bar{y}(s)$ from (4) & (5) we get

$$\bar{x} = \frac{1}{2}\left\{\frac{1}{s-1}\right\} + \frac{1}{2}\left\{\frac{s}{s^2+1}\right\} + \frac{1}{2}\left\{\frac{1}{s^2+1}\right\} + \frac{1}{(s^2+1)^2}$$

$$L^{-1}\{\bar{x}(s)\} = \frac{1}{2}L^{-1}\left\{\frac{1}{s-1}\right\} + \frac{1}{2}L^{-1}\left\{\frac{s}{s^2+1}\right\} + \frac{1}{2}L^{-1}\left\{\frac{1}{s^2+1}\right\} + L^{-1}\left\{\frac{1}{(s^2+1)^2}\right\}$$

$$x(t) = \frac{1}{2}e^t + \frac{1}{2}\cos t + \frac{1}{2}\sin t + \frac{1}{2}(\sin t - t\cos t)$$

$$= \frac{1}{2}e^t + \frac{1}{2}\cos t + \sin t - \frac{t}{2}\cos t \quad \text{——} (6)$$

using (6) in (1), we get

$$\boxed{y(t) = -\frac{1}{2}e^t - \frac{1}{2}\sin t + \frac{1}{2}\cos t + \frac{t}{2}\sin t}$$

Q 3

a) Find the fourier transform of $f(x) = \begin{cases} 1-x^2 & |x| \leq 1 \\ 0 & |x| > 1. \end{cases}$

→ The fourier transform of $f(x)$ is given by

$$F\{f(x)\} = \bar{F}(s) = \int_{-\infty}^{\infty} f(x)\, e^{isx}\, dx$$

$$= \int_{-\infty}^{-1} f(x) . e^{isx} dx + \int_{-1}^{1} f(x)\, e^{isx} dx + \int_{1}^{\infty} f(x) e^{isx}\, dx$$

$$= \int_{1}^{1} (1-x^2) e^{isx} dx$$

$$= \left| (1-x^2) \frac{e^{isx}}{is} - (-2x) \frac{e^{isx}}{i^2 s^2} + (-2) \frac{e^{isx}}{i^3 s^3} \right|_{-1}^{1}$$

$$= (0-0) + 2(1) \frac{e^{is}}{i^2 s^2} - 2(-1) \frac{e^{-is}}{i^2 s^2} - 2 \frac{e^{is}}{i^3 s^3} + \frac{2e^{-is}}{i^3 s^3}$$

$$= 2\left[\frac{e^{is}}{-s^2} + \frac{e^{-is}}{-s^2} - \frac{e^{is}}{-is^3} + \frac{e^{-is}}{-is^3} \right]$$

$$= 2\left[\frac{e^{is} + e^{-is}}{-s^2} - \left(\frac{e^{is} - e^{-is}}{-is^3} \right) \right]$$

$$= 2\left[-\frac{2\cos s}{s^2} + \frac{2\sin s}{s^3} \right]$$

$$= \frac{-4}{s^3} [s\cos s - \sin s)$$

Ans.

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

Q3b) Find the fourier sine transform of $e^{-|x|}$. Hence show that $\int_0^\infty \frac{x\sin mx}{1+x^2}\,dx = \frac{\pi e^{-m}}{2}$ $m>0$.

$\rightarrow$ $f(x) = e^{-|x|}$

fourier sine transform

$$f_s(s) = \int_0^\infty f(x)\sin sx\,dx$$

Inverse fourier sine transform

$$f(x) = \frac{2}{\pi}\int_0^\infty F_s(s)\sin sx\,ds.$$

$$F_s(s) = \int_0^\infty e^{-x}\sin sx\,dx$$

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$$

$a=-1$

$b=s$

$$F_s(s) = \left[\frac{e^{-x}}{1+s^2}(-\sin sx - s\cos sx)\right]_0^\infty \qquad \boxed{e^{-\infty}=0}$$

$$= \frac{0-(-0-s)}{1+s^2} = \frac{s}{1+s^2}$$

Inverse fourier sine transform

$$f(x) = \frac{2}{\pi}\int_0^\infty F_s(s)\sin sx\,ds$$

$$e^{-x} = \frac{2}{\pi}\int_0^\infty \frac{s}{1+s^2}\sin sx\,ds$$

Change x to m

$$e^{-m} = \frac{2}{\pi}\int_0^\infty \frac{s}{1+s^2}\sin sm\,ds$$

$$\rightarrow \pi.\frac{e^{-m}}{2} = \int_0^\infty \frac{s\sin ms}{1+s^2}\,ds$$

or

$$\pi.\frac{e^{-m}}{2} = \int_0^\infty \frac{s\cdot\sin ms}{1+s^2}\,ds$$

replac s by x

$$= \int_0^\infty \frac{x\sin mx}{1+x^2}$$

Q3) c) Using Parseval's Identity prove that.

$$\int_0^{\infty} \frac{t^2}{(t^2+1)^2}\, dt = \frac{\pi}{4}$$

$\rightarrow$ consider $f(x) = \dfrac{x}{x^2+1}$ then $F_s(s) = \dfrac{\pi}{2} \cdot e^{-s}$

use Parsevals identity for fourier sine transformation

$$\rightarrow \frac{2}{\pi} \int_0^{\infty} [F_s\{f(x)\}]^2\, ds = \int_0^{\infty} [f(x)]^2\, dx$$

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$$= \frac{2}{\pi} \int_0^{\infty} \left(\frac{\pi}{2} e^{-s}\right)^2 ds = \int_0^{\infty} \frac{x^2}{(x^2+1)^2}\, dx.$$

$$= \frac{2}{\pi} \int_0^{\infty} \left(\frac{\pi}{2} (e^{-s})\right)^2 ds = \frac{\cancel{2}}{\cancel{\pi}} \cdot \frac{\pi^{\cancel{2}}}{\cancel{4}} \int_0^{\infty} e^{-2s}\, ds$$

$$= \frac{\pi}{2} \left[\frac{e^{-2s}}{-2}\right]_0^{\infty}$$

$$= \frac{\pi}{-4} (e^{-\infty} - e^{-0}) = \boxed{\pi/4}$$

$$\int_0^{\infty} \frac{t^2}{(t^2+1)^2}\, dt = \frac{\pi}{4}.$$

//

→ Find the temperature in a bar of length 2 units whose

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1) $Pp + Qq = R$

2) AE

$$\frac{\partial x}{P} = \frac{\partial y}{Q} = \frac{\partial x}{R}$$

Q 4)

a) Solve the Partial differential equtions

$$(x^2 - yz)p + (y^2 - zx)q = z^2 - xy.$$

Solu) $(x^2-yz)p + (y^2-2x) = z^2-xy$

is of the form Lagranges Linear PDE

$$AE \quad \frac{\partial x}{x^2-yz} = \frac{\partial y}{y^2-zx} = \frac{\partial z}{z^2-xy}$$

Now

$$\frac{dx-dy}{x^2-yz-y^2+zx} = \frac{dy-dz}{y^2-zx-z^2+xy} = \frac{dz-dx}{z^2-xy-x^2+yz}$$

$$\frac{dx-dy}{(x^2-y^2)+z(x-y)} = \frac{\partial y-dz}{(y^2-z^2)+x(y-z)} = \frac{dz-dx}{(z^2-x^2)+y(z-x)}$$

$$\rightarrow \frac{dx-dy}{(x-y)\,(\cancel{x+y+z})} = \frac{dy-dz}{(y-z)\,(\cancel{y+z+x})} = \frac{dz-dx}{(z-x)\,(z+x+y)}$$

$$\rightarrow \frac{\partial x-\partial y}{(x-y)} = \frac{\partial y-dz}{(y-z)} = \frac{\partial z-dx}{(z-x)}$$

$$\rightarrow \frac{\partial x}{x-y} - \frac{\partial y}{x-y} = \frac{\partial y}{y-z} - \frac{dz}{y-z} =$$

Integrate on both sides

$$\log(x-y) - \log(x-y) = \log(y-z) - \log(y-z)\,(-1) + \log c_1$$

$$2\log(x-y) = 2\log(y-z) + \log c_1$$

$$2\log\left[\frac{x-y}{y-z}\right] = \log c_1$$

$$\log\left[\frac{x-y}{y-z}\right]^2 = \log c_1$$

$$\left(\frac{x-y}{y-z}\right)^2 = c_1 \rightarrow \frac{x-y}{y-x} = \sqrt{c_1} = c_2$$

Take last 2 terms.

$$\frac{\partial y}{y-z} - \frac{\partial z}{y-z} = \frac{\partial z}{z-x} - \frac{\partial x}{z-x}$$

integrate on both sides

$$\log(y-z) - \log(y-z)\,(-1)$$

$$= \log(z-x) - \log(z-x)\,(-1) + \log c_3$$

Q4)

$$2\log\left(\frac{y-2}{2-x}\right) = \log c_3$$

$$\left(\frac{y-2}{2-x}\right)^2 = c_3$$

$$\frac{y-x}{2-x} = \sqrt{c_3} = c_4$$

Ans.

2018

Qu.

b) Use method of separation of variables to solve the equation $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$; given that $u(x,0) = 6e^{-3x}$.

$\rightarrow$ Assume the solution to be

$u(x,t) = X(x)\,T(t)$ —— ②

X is a function of x only

T is a function of t only

Diff. eqⁿ ② w.r.t. x partially.

$\frac{\partial u}{\partial x} = \frac{\partial X}{\partial x}\cdot T$ —— ③

Diff. eqⁿ ② w.r.t. t partially.

$\frac{\partial u}{\partial t} = \frac{\partial T}{\partial t}\cdot X$ —— ④

from ①, ③ and ④

$\frac{\partial X}{\partial x}\cdot T = 2\frac{dT}{dt}\cdot X + X\cdot T$

Dividing by XT throughout.

$\frac{dX}{dx}\cdot\frac{1}{X} = 2\frac{dT}{dt}\cdot\frac{1}{T} + 1 = K$ (constant)

$\frac{dX}{dx}\cdot\frac{1}{X} = K$

$\int\frac{dX}{X} = \int K\,dx$

$\log X = Kx + \log c_1$

$\log X - \log c_1 = Kx$

$\log\left(\frac{X}{c_1}\right) = Kx$

$\frac{X}{c_1} = e^{Kx}$

$X = c_1 e^{Kx}$ —— ⑤

$2\frac{dT}{dt}\cdot\frac{1}{T} + 1 = K$

$2\frac{dT}{dt}\cdot\frac{1}{T} = K-1$

$2\int\frac{dT}{T} = \int(K-1)\,dt$

$2\log T = (K-1)t + \log c_2$

$\log T^2 = (K-1)t + \log c_2$

Qu 0) Find the temperature in a bar of length 2 units, whose ends are kept at zero temperature and lateral surface insulated if the initial temperature is $\sin\frac{\pi x}{2}$ ~~x~~ $3\sin\frac{5\pi x}{2}$.

$\Rightarrow$ one-dimensional heat equation is

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{---} (1)$$

The solution of equation (1) consistent with the physical nature of the problem is given by

$$u = c_1 e^{-m^2c^2t}(c_2\cos mx + c_3\sin mx) \quad \text{---} (2)$$

where $u(x,t) = 0$ at $x=0$ ——— (3)

$u(x,t) = 0$ at $x=2$ ——— (4)

$u(x,t) = \sin\frac{\pi x}{2} + 3\sin\frac{5\pi x}{2}$ at $t=0$ ——— (5)

using the condition (3) in (2) we obtain

$$0 = c_1 e^{-m^2c^2t}(c_2)$$

$$\Rightarrow c_2 = 0$$

from (2) we have therefore

$$u = c_1 c_3 e^{-m^2c^2t}\sin mx \quad \text{---} (6)$$

using condition (a) in (2) we obtain

$$0 = c_1 c_3 \bar{e}^{m^2 c^2 t} \sin 2m$$

$$\Rightarrow \sin 2m = 0 = \sin n\pi$$

$$\Rightarrow 2m = n\pi$$

$$\Rightarrow m = \frac{n\pi}{2}$$

from (6) we have there fore

$$u = c_1 c_3 \bar{e}^{(\frac{n\pi}{2})^2 c^2 t} \sin \frac{n\pi x}{2} \quad \text{——— (7)}$$

The general solution is

$$u = \sum_{n=1}^{\infty} b_n \bar{e}^{(\frac{n\pi}{2})^2 c^2 t} \sin \frac{n\pi x}{2} \quad \text{——— (8)}$$

using (5) in (8) we obtain

$$\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2} = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{2}$$

$$= b_1 \sin \frac{\pi x}{2} + b_2 \sin \frac{2\pi x}{2} + b_3 \sin \frac{3\pi x}{2} + b_4 \sin \frac{4\pi x}{2} + b_5 \sin \frac{5\pi x}{2} + \ldots$$

$$\Rightarrow b_1 = 1, \; b_5 = 3, \; b_2 = b_3 = b_4 = b_6 = b_7 = b_8 = 0$$

from (8) we have there fore

$$u = b_1 e^{\frac{-\pi^2}{4} c^2 t} \sin \frac{\pi x}{2} + b_5 e^{-(\frac{5\pi}{2})^2 c^2 t} \sin \frac{5\pi x}{2}$$

$$\text{i.e } u = e^{\frac{-\pi^2}{4} c^2 t} \sin \frac{\pi x}{2} + 3 \bar{e}^{(\frac{5\pi}{2})^2 c^2 t} \sin \frac{5\pi x}{2},$$

which is the required temperature.

Q 5

a) If $f(z)$ is analytic function with constant modulus show that $f(z)$ is constant.

→ proof : Let $f(z) = u+iv$ be an analytic function with const. modulus.

To prove : $f(z)$ is const.

$f(z)$ is analytic

→ CR condition is satisfied

i.e $u_x = v_y$ —— ① $u_y = -v_x$ —— ②

Now we have

$|f(z)| = \text{const.}$

$\rightarrow \sqrt{u^2+v^2} = \text{const.}$

squring b.s we ge

$(u^2+v^2) = \text{const}$ —— ③

diff. ③ w.r.t. x we get

$\rightarrow 2u \frac{du}{\partial x} + 2v \frac{\partial v}{\partial x} = 0$

$2\left[u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x}\right] = 0$

$\rightarrow u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x} = 0$ —— ④

diff ③ w.r.t. y

$2u \frac{\partial u}{\partial y} + 2v \frac{\partial v}{\partial y} = 0$

$u \frac{\partial u}{\partial y} + v \frac{\partial v}{\partial y} = 0$ —— ⑤

using CR condition ① & ② in eq^n ⑤ & ④

$u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x} = 0$ —— ⑤

$-u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = 0$ —— ⑥

multiply ⑤ by u & ⑥ by v & adding we get

$u^2 \frac{du}{\partial x} + uv \frac{\partial v}{\partial x} + \left[uv \frac{\partial v}{\partial x} + v^2 \frac{\partial v}{\partial y}\right] = 0$

$u^2 \frac{\partial u}{\partial x} + v^2 \frac{\partial v}{\partial y} = 0$

$u^2 \frac{du}{\partial x} + v^2 \frac{\partial u}{\partial x} = 0$ —— [using eq^n ①]

$(u^2+v^2) \frac{\partial u}{\partial x} = 0$

$\sqrt{u^2+v^2} = \text{const}$

$u^2+v^2 = \text{const}^2$

(6) $\oint \frac{}{c\,(z}$

Q b) Evalute $\oint \tan z \, dz$, where c is the circle $|z| = 2$

→ ~~$f(z) = \sin$~~

$f(z) = \tan z$, $|z| = 2$

$f(z) = \frac{\sin z}{\cos z} = \frac{\phi(z)}{\psi(z)}$,

$f(z) = \frac{\phi(z)}{\psi(z)}$

If at $z = z_0$, $\psi(z) = 0$ and $\phi(z) \neq 0$

$\psi'(z_0) \neq 0$ then z_0 would be a simple pole

$$\text{Residue} = \frac{\phi(z_0)}{\psi'(z_0)}$$

$f(z) = \tan z$; $\phi(z) = \sin z$, $\psi(z) = \cos z$.

$\cos z = 0$ for $z = (2n+1)\frac{\pi}{2}$

$|z| = 2$, only $z = \pm\frac{\pi}{2}$ points will fall within the given circle c.

$$\therefore \text{Residue at } \left(z = \frac{\pi}{2}\right) = \frac{\sin \pi/2}{-\sin \pi/2} = -1$$

$$\text{Residue at } z = -\pi/2 = \frac{\sin(-\pi/2)}{-\sin(-\pi/2)} = -1$$

$\oint_c \tan z \, dz$ whitin $|z| = 2$

$$\boxed{= 2\pi i(-1-1) = -4\pi i}$$

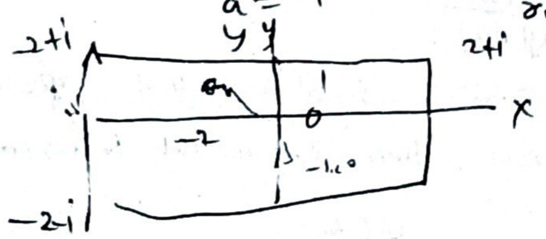
Q.6 using Cauchy's integral formula

Q a);

1) $\oint \frac{\cos(\pi z)}{(z^2-1)} dz$ around a rectangle with vertices $2 \pm i, -2 \pm i$

→ a) Here $f(z) = x$ ~~...~~ and $f(z) = \cos \pi z$ is analytic in the region bounded by the given rectangle.

The two non-singular point $a = 1$ and $a = -1$ also lie insid this rectangle.



Now ~~$\oint \frac{\cos \pi z}{(z^2-1)} dx = \frac{1}{2} \oint_{C_1} \frac{\cos \pi}{}$~~

Now $\oint_C \frac{\cos \pi z}{z^2-1} dz = \frac{1}{2} \oint_C \frac{\cos \pi z}{z-1} dz - \frac{1}{2} \oint \frac{\cos \pi z}{z+1} dz$

$= \frac{1}{2}[2\pi i f(1) - 2\pi i f(-1)]$ — By Cauchy Integral

$= \pi i [f(1) - f(-1)] = \pi i [\cos \pi - \cos(-\pi)]$

$= \pi i [-1+1] = 0$

b) $\oint_C \frac{\sin^2 z}{(z-\frac{\pi}{6})^3} dz$ where C is the circle $|z|=1$

→ The function $f(z) = \sin^2 z$ is analytic inside and on the circle $C: |z|=1$ and the singular point $a = \frac{\pi}{6}$ lies within C, Hence, by Cauchy, integral Formula

$$f''(a) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^3} dz$$

$$\therefore \oint_C \frac{\sin^2 z}{(z-\frac{\pi}{6})} dz = \pi i \left[\frac{d^2}{dz}(\sin^2 z)\right]_{z=\frac{\pi}{6}} = \pi i [2\cos 2z]_{z=\frac{\pi}{6}}$$

$$= \pi i \left[2 \cos \frac{\pi}{3}\right] = \pi i.$$